

Hybrid approach of Ant Colony system and Recurrent Neural network for Energy Efficient Routing in Cognitive Wireless Sensor Networks

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Abstract—Energy-aware communication in Cognitive Wireless Sensor Networks (CWSNs) was challenging with dynamic spectrum availability, high topology change rates, and low energy. Conventional routing techniques result in excessive energy usage and reduced network lifetimes. This work introduces an adaptive AI hybrid technique integrating Enhanced Ant Colony Optimization (EACO) for desirable choices in CH and sequence prediction neural network for energy pattern prediction and proactive optimization. The Decentralized Power-Aware Clustering (DPAC) protocol also optimizes energy distribution. Regarding energy efficiency, mean throughput, and network lifetime, our model outperformed existing systems. Simulations show that EACO + RNN greatly reduces energy use, improves the reliability of packet transmission, and extends the network's life by stopping early node failures. This hybridization of predictive AI and bio-inspired optimization guarantees scalable, energy-efficient CWSNs.

Keywords—Cognitive Wireless Sensor Networks (CWSNs), Energy-Efficient Routing, Enhanced Ant Colony Optimization (EACO), Recurrent Neural Network (RNN), Decentralized Power-Aware Clustering (DPAC), hybrid AI approach, adaptive routing, network lifetime optimization, bio-inspired optimization, machine learning in WSN.

I. INTRODUCTION

Cognitive Wireless Sensor Network (CWSN) is an evolved solution to overcome the shortcomings of traditional Wireless Sensor Networks (WSNs) by facilitating dynamic spectrum access and smart data transmission. They are networks comprising sensor nodes that are designed to function within an energy-constrained setting and thus need to be equipped with efficient routing strategies to extend the network lifespan and guarantee stable communication. Energy expenditure in CWSNs is primarily determined by data transmission, reception, and computational processing. Conventional routing protocols do not adjust to the varying energy levels of sensor nodes, leading to fast node energy depletion and network partitioning.

This results in an inferior network lifetime, lower packet delivery ratios, and increased latency, making energy-efficient routing a pressing research priority. Some current energy-aware routing strategies focus on minimizing energy expenditure using clustering methodologies. These protocols are not responsive to dynamic variations in energy levels and fail to effectively apply learning-based prediction methodologies for energy-conscious decision making.

In response to these challenges, this study recommends an adaptive AI hybrid solution that combines Enhanced Ant Colony Optimization (EACO) and Recurrent Neural Network (RNN) for energy-efficient routing in CWSNs. The basic idea of this solution is to merge the optimization power of swarm intelligence algorithms with the predictability of deep learning models to optimize the routing efficiency. The proposed framework has three components: (i) EACO for cluster head selection and routing, where optimization using pheromones assists in finding energy-efficient paths; (ii) LSTM-based RNN energy forecasting, which identifies historical patterns in energy consumption and avoids proactively choosing energy-low nodes for data transmission; and (iii) the DPAC protocol, which enables load balancing in the network by dynamically adjusting the head node in relation to the available power levels. The foremost aim of the EACO + RNN hybrid is to reduce power consumption, promote data transfer performance, and maximize the network reliability. The major contributions of this research are energy-aware routing optimization, adaptive energy forecasting, dynamic cluster head election, comparative study with other models such as ICOA (Improved Chaos Optimization Algorithm) and IJO-LF (Improved Jaya Optimization-Levy Flight), and MATLAB-based simulations, confirming the superior performance of our method.

The combination of EACO and LSTM is a new method for addressing energy-efficiency issues in CWSNs. In contrast to conventional clustering methods, the adaptive AI-based

framework learns dynamically from network conditions and adapts routing decisions in real-time. The integration of swarm intelligence (EACO) and deep learning (LSTM) provides improved energy-aware routing and is a potential solution for future wireless sensor networks. This work opens avenues for extended progress in hybrid AI-optimized protocols, specifically on IoT-based sensor networks, intelligent city deployments, and industrial surveillance systems, where power efficiency is the most important. In addition, the proposed methodology can be generalized to facilitate a heterogeneous network infrastructure and multi-goal optimization frameworks with enhanced efficiency across intelligent wireless networks.

II. RELATED WORKS

Zhang, Bin, et al., an effective Cooperative Spectrum Sensing (CSS) algorithm to maximize reporting time and increase throughput in Cognitive Wireless Sensor Networks (CWSNs). Nonetheless, this method may suffer from high computational complexity and possible loss of valuable sensing information owing to selective reporting and the requirement of accurate coordination among sensor nodes and the fusion center, which would affect real-time decision-making in large-scale systems [1]. The goal of Valehi, Ali, and Abolfazl Razi's research is to reduce CWSN power usage. This study optimizes the data transmission performance by determining the optimal number of samples to aggregate into packets and examining the queuing behavior, packet reject rates, and retransmission probabilities. [2]. Most methods, particularly those AI and metaheuristic-based methods, have been studied to ensure maximum energy efficiency for WSN. Kumar and Singh [3], for instance, presented a hybrid method of dynamic clustering by integrating genetic algorithms (GA) and fuzzy logic. Although their method effectively minimized energy consumption, it lacked adaptability in spectrum conditions, which is a required feature of cognitive WSNs.

A PSO-RNN approach was presented in [4] that used Particle Swarm Optimization to choose the optimal CH and a Recurrent Neural Network for predicting traffic. Routing responsiveness was improved, but computational costs were high and proactive spectrum adaptability did not exist. Similarly, Deep Q-Networks (DQNs) were used in [5] to achieve dynamic route selection, but the model complexity.

In [6], Ant Colony Optimization (ACO) was implemented in combination with reinforcement learning in order to cluster formation optimization. Nevertheless, predictive modeling for energy dynamics was not included. [7] implemented node status estimation by machine learning-based classifiers but did not present a solution towards the problem of non-stationary spectrum availability in CWSNs.

Srividhya, V., and T. Shankar, the authors, aim to enhance bandwidth usage for power awareness in CWSN by proposing Low-Energy Proximity-Based Frequency-Aware Enhancement (LPFE) technique. The method uses Bee Colony Optimization (BCO) for cluster establishment and Donkey Intelligence Algorithm (DIA) for power-saving communication to extend system reliability. But this approach can be challenged by massive computations involved, the need for precision lookup tables in routing to ensure accuracy, and perhaps limitations dynamic channel change where the spectrum bandwidth and network topology keep varying [8].

Phung-Duc et al. and Tuan et al.. However, there are challenges such as possible delays owing to repeated attempts at sensing, added system complexity, and decreased UU transmission efficiency in high-traffic environments with frequent LU

interruptions [9,10]. Valaskova, K.; Horak, J.; Lăzăroiu, G. [11] examines socially responsible technologies in autonomous mobility. It emphasizes how cognitive wireless sensor networks and virtual data modeling assist autonomous systems. The article addresses legal and ethical points of applying such technologies. [12] Andronic, M.; IoT sensing networks, artificial intelligence decision-making, and deep learning for intelligent process management in cyber-physical manufacturing systems. It facilitates in Industry 4.0 environments more intelligent, more efficient industrial processes.[13] Wang, Z., then colleagues. an energy-efficient cluster-based routing algorithm for Wireless Sensor Networks (WSNs). For optimal routing it hybridizes Ant Colony Optimization and Firefly Algorithm. Kannappan, A.; Bommi, R. M.[14] Energy-efficient routing a novel hybrid Bilevel-Litechenberg Optimization Algorithm with Ant Colony Optimization. Yang, Y.; Li, Y.; Wang, T.; & Liu, G. [14] a deep reinforcement learning (DRL)-based method towards improving DCN energy efficiency. Without sacrificing performance, the method manages traffic and routing dynamically to conserve power. The model learns to adapt to varying network conditions for efficient energy consumption through DRL. [15] a vessel trajectory prediction model based on LSTM neural network tang, H., Yin, Y., & Shen, H. In making accurate predictions, the model predicts temporal correlations in marine vessel motions. It assists in improving navigation efficiency and marine safety.

On the basis of a hybrid African Vultures-Cuckoo Search Optimization algorithm, Asha, A., Verma, N., and Poonguzhali, I. [16] an energy-efficient routing protocol for WSNs. The method is for multi-objective optimization with the goal to achieve a balance between network performance and energy consumption. It improves reliability and lifespan of nodes during data transmission. An energy-aware, depth-regulated routing system for UWSNs, Lilhore, U. K., et al. [17] The system relies on depth information to enhance routing decision-making and reduce energy wastefulness. The system ensures low power low data transmission efficiency. Ali Forghani Elah, Abadi, et al. Through reinforcement learning to develop a very power-aware control and path selection procedure, the author is optimistic about enhancing the lifespan of WSN. This method restricts data transmission based on the rate of change of data, maximizes routing, and employs sleep scheduling to save energy[18].

Comparative Analysis of Hybrid AI-Metaheuristic Routing Methods in WSNsHybrid models merging metaheuristic optimization with machine learning models have also drawn interest in WSNs for handling energy efficiency and adaptability. Despite their popularity, the majority of existing methods suffer from trade-offs in learning accuracy, convergence rate, or adaptability in the face of dynamic spectrum.

III. EXISTING SYSTEM

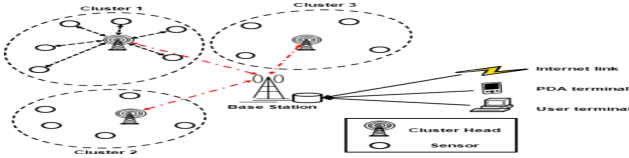
The existing systems in Cognitive Wireless Sensor Networks (CWSNs) rely primarily on conventional optimization techniques such as Improved Chaotic Optimization Algorithm (ICOA) and Improved Jaya Optimization with Levy Factor (IJO-LF). Even though these techniques tend to promote energy efficiency, they are confronted by serious challenges that include dynamic availability of spectrum, large topology change rates, and non-uniform distribution of energy across sensor nodes. Conventional cluster-based protocols, such as LEACH and DPAC, can somewhat increase the network

lifetime, but they are insensitive to the fluctuating energy consumption in real time, which results in unfair energy use and node failures too soon. Most of the existing systems also do not have predictive modeling incorporated, thereby making the routing. It leads to inefficient energy consumption, heightened data loss, and more network congestion. The absence of an intuitive method to reoptimize routes for routing based on future patterns of energy consumption also limits the efficiency of these methods. Therefore, existing solutions are marred by wasted energy at large scales, reduced packet transmission reliability, and an overall reduced lifespan for the network. Without adaptive learning and proactive optimization, current routing mechanisms cannot ensure lower power usage and system throughput, which makes the necessity of more advanced AI-based hybrid solution more critical.

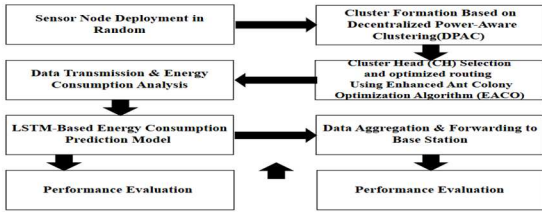
IV. PROPOSED SYSTEM

The Cognitive Wireless Sensor Network (CWSN) is attained through the combination of an AI-based adaptive routing technique. This is accomplished through deploying sensor nodes, cluster formation, optimal selection of cluster heads (CHs), routing path optimization, and utilization of AI-based energy prediction for efficient network performance. In our approach, Enhanced Swarm Intelligence Optimization and Recurrent Neural Network are combined with Decentralized Power-Aware Clustering (DPAC) to improve energy efficiency.

A. Network Structure



B. Flow Graph



C. Sensor Node Deployment

Nodes are initially placed randomly across the target area to form a CWSN. The nodes are tasked with forwarding the gathered data and monitoring environmental conditions. Licensed Users (LUs) and Unlicensed Users (UUs) are the two different types of users on the network. While LUs are allotted preference in utilizing the spectrum, secondary users access available spectrum while avoiding interference with licensed users (LUs). Because of this aspect, proper spectrum management techniques need to be employed so that both types of users can have fair utilization. Using advanced algorithms, the network can dynamically reallocate resources, achieve performance optimization, and provide seamless communication among nodes. Sensing agents are arbitrarily deployed in the given field size for providing even coverage over the area. Energy level, position, and communication range are initialized to every node. The Gateway was present in the network to receive aggregated data send by the sensing agents.

D. Cluster Head Selection using EACO

An improved method named Enhanced Ant Colony Optimization (EACO) with adaptive pheromone update and heuristic functions enhances classical ACO to elect head nodes optimally in CWSN. Because head nodes aggregate data agents report to the gateway, an appropriate CH selection is mandatory for increase network lifetime and power conservation. By adjusting pheromone evaporation rates dynamically and incorporating energy-aware heuristic procedures, EACO enhances ACO and provides load balancing and a larger network lifetime. Following ant foraging behavior, EACO proposes adaptive pheromone updates according to energy depletion patterns, node connectivity, and traffic load. Every node is considered an ant, and its chances of being a CH are calculated based on a more enhanced fitness function:

$$P_i = \frac{(\tau_j^\alpha)(\eta_j^\beta)(E_j^\gamma)}{\sum_{j \in N} (\tau_j^\alpha)(\eta_j^\beta)(E_j^\gamma)} \quad (1)$$

where τ is the pheromone level, η is the heuristic information (e.g., remaining energy), α and β are the tuning parameters and E is the lasting power of the agents. The EACO-based CH choosing avoid nodes with lesser energy levels, poor location, and high communication cost as CHs. Also, EACO proposes a hybrid pheromone update rule to intensify high-energy, low-latency routes while penalizing unproductive routes. This energy-driven CH selection increases network lifetime, decrease in energy utilization, and enhancement in load balance among sensor agents.

E. Cluster formation using DPAC

Once CHs are selected by EACO, cluster formation takes place. DPAC protocol is used to form clusters dynamically based on energy. The nodes calculate their joining probability to a cluster in terms of range from the CH and available power. The joining probability of nodes to chosen as a CH is calculated as:

$$P_i = \frac{E_i}{\bar{E}_{network}} \quad (2)$$

where $\bar{E}_{network}$ is the average power value of all agents, E_i is the limiting energy of agent i , and P_i is the probability that agent i will be detected as a CH.

This traditional DPAC model is contextually expanded to fit within the predictive paradigm facilitated by the LSTM module. Instead of basing clustering decisions on momentary energy levels, the fusion with LSTM predictions brings in a dynamic weighting scheme. More particularly, nodes predicted to experience considerable energy drops (as predicted by LSTM based on past usage patterns) are given a lesser chance of acting as cluster heads, even if their present energy E_i seems adequate. This proactive adjustment reduces the possibility of choosing nodes that will get exhausted in the near future, thus improving cluster stability in the long run. Furthermore, DPAC's decentralized decision-making is maintained to guarantee scalability, but its responsiveness to future energy shortages introduces a new layer of intelligence not found in traditional applications of the protocol. By this hybrid alignment, DPAC serves not just as a clustering mechanism but as an energy-aware real-time coordination layer augmenting the synergy between EACO-based optimization and AI-driven prediction.

F. Packet Transfer and Power Consumption Analysis

After clusters are created, detector agents use either single-hop multi-hop routing, depending on the EACO-optimized path, to deliver their data to designated Cluster Heads (CHs). CHs collect information and shoot the gateway the compressed

interpretation. The energy used for transmission and event is tracked and recorded throughout the process.

$$E_{tx} = E_{elec} \times k + E_{amp} \times k \times r^2 \quad (3)$$

where E_{tx} is the power usage for data transfer, E_{elec} is the power dissipation per bit, E_{amp} is the amplification energy, k is packet size, and r is the range from sender to receiver. Likewise, the energy spent in reception is illustrated as:

$$E_{rx} = E_{elec} \times k \quad (4)$$

where E_{rx} is energy needed to receive data. These models are employed to monitor per-node energy depletion, allowing real-time routing adjustments. As the energy level decreases, the LSTM-based prediction model predicts future depletion patterns, and nodes with higher drain rates are skipped in the next rounds of CH selection or routing. This combined feedback loop prevents premature node failure from energy depletion and guarantees an even distribution of energy consumption throughout the network.

G. LSTM-Based Energy Consumption Prediction

To forecast energy consumption in sensor nodes, an LSTM neural network is employed to allow proactive routing and premature energy-critical node avoidance. Training was conducted using a simulated dataset of 10,000 time-series samples where each sample consists of a series of 10 past time steps that have features like residual energy, transmission load, cluster head (CH) history, and neighborhood density. The model architecture consists of two 64- and 32-unit LSTM layers, a fully connected output layer for single-value energy prediction, and a dropout layer with a drop rate of 0.2 to avoid overfitting. With an 80:20 train-validation split and a batch size of 32, the model was trained over 100 epochs using the Adam optimizer and MSE loss function. A Root Mean Square Error (RMSE) of 0.018 during validation indicates that the model performed well in terms of generalization across various node behaviours. The LSTM model operates by processing input features via forget, input, and output gates, which regulate information flow in the network.

The discard gate determines which past energy values should be retained or discarded using the formula

$$d_t = \sigma[W_d \times [h_{t-1}, X_t] + b_d] \quad (5)$$

where d_t is the discard gate activation, x_t represents energy consumption data, h_{t-1} is the hidden value of previous state, W and b are weights and bias, respectively.

The update gate updates the cell state using,

$$u_t = \sigma[W_u \times [h_{t-1}, X_t] + b_u] \quad (6)$$

$$\sim S_t = \tanh[W_s \times [h_{t-1}, X_t] + b_s] \quad (7)$$

where U_t is the update gate activation and C_t is the internal memory cell.

The cell state update is then computed as

$$S_t = f_t \times S_{t-1} + i_t \times \sim S_t \quad (8)$$

where S_t represents long-term energy trends.

The visibility gate determines the predicted energy level using

$$V_t = [W_v \times [h_{t-1}, X_t] + b_v] \quad (9)$$

$$h_t = V_t \times \tanh(S_t) \quad (10)$$

$$E_{(t+1)} = W_y \times h_t + b'_y \quad (11)$$

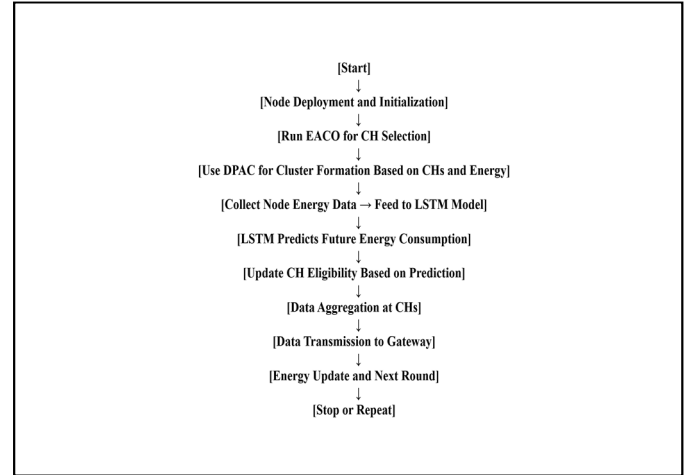
where V_t controls the output information, h_t represents the predicted energy value, $E_{(t+1)}$ denotes the predicted energy of the nodes.

The discard gate chooses what previous data should remember or ignore, the input gate provides new energy consumption values to update the internal memory cell, and the visibility gate computes node's predicted future energy level.

The LSTM model used in this research is tailored to forecast future energy consumption of a single sensor node using

temporal patterns. The model has two hidden LSTM layers with 64 units each and a dense output layer for forecasting the residual energy at the subsequent time step. Normalized features like past energy levels, packet transmission rate, cluster density, and distance to CH are fed as input to the model. It was trained on a synthetically created dataset from simulation logs from 500 nodes for 50 rounds, resulting in 25,000 training samples. The short-run patterns in energy consumption were embodied with a length of ten steps of time. A training dataset of 80% and a validation dataset of 20% were formulated. With 0.001 as the learning rate, the model was optimized with the Adam optimizer, the loss function being Mean Squared Error (MSE). To prevent overfitting, the network was trained for 100 epochs, where it was early stopped according to validation loss. To influence CH selection and routing in the next round actively, this LSTM model runs locally within the simulation and refreshes predictions at the end of each round.

H. Flowchart of hybrid approach



V. RESULT AND DISCUSSION

The experiment was conducted with a **MATLAB**-oriented environment specifically designed to simulate Cognitive Wireless Sensor Networks (**CWSNs**). A **500m × 500m** two-dimensional sensor square field was established, in which 500 static **sensor nodes** were dispersed randomly. The nodes are immobile during the simulation in order to concentrate on measuring routing efficacy without interference caused by mobility.

One gateway node was placed in the middle of the field to receive aggregated data from CHs. A starting energy of 2 Joules was assigned to each node, and a first-order radio model was used for packet transmission.

The cognitive abilities of every node were simulated with a binary spectrum sensing scheme. Channels were modeled to follow an ON-OFF Markov process, in which idle (available) and busy (occupied) states were probabilistically determined. Idle states supported data transmission, whereas busy states necessitated spectrum switching. Spectrum availability was dynamically updated per round to adapt to real-time sensing and environmental fluctuation.

All simulation performance measures—i.e., residual energy, CH stability, and packet delivery—were computed across 20 independent runs to promote statistical consistency and minimize the influence of random variability inherent in stochastic models such as EACO. Each result plotted here is the average of 20 simulation runs, providing consistency and

reducing randomness associated with metaheuristic-based routing.

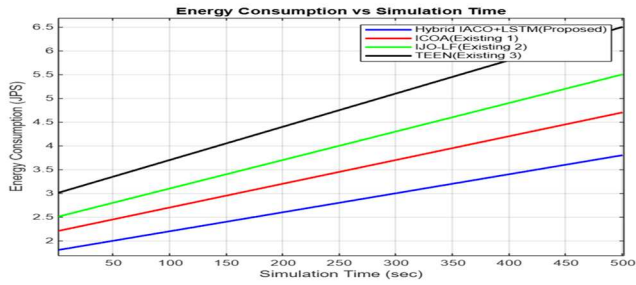


Fig. 1. Energy Consumption (JPS) VS Simulation Time (Sec)

Fig. 1 depicts the proposed EACO+LSTM-based routing model for Cognitive Wireless Sensor Networks (CWSNs). It displays the interactions between the sensing nodes, cluster heads, and LSTM predictor.

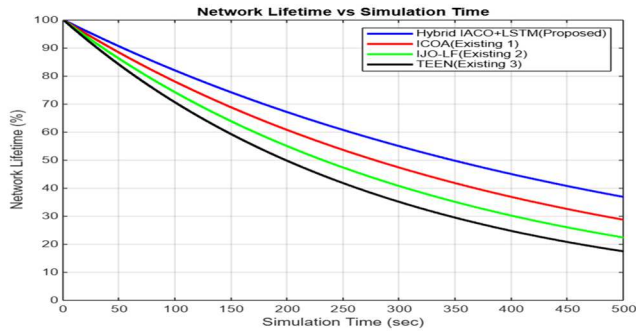


Fig. 2. System Reliability (%) VS Simulation Time (Sec)

The flowchart of the proposed hybrid EACO+LSTM routing mechanism is displayed in Fig. 2. It describes the decision-making process that starts with data sensing and moves on to CH selection via EACO, LSTM prediction, and route establishment.

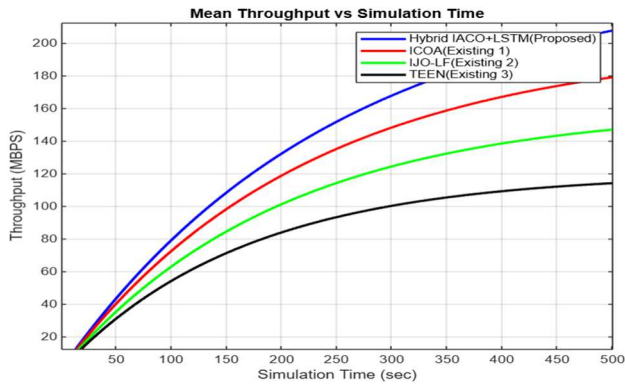


Fig. 3. Mean Throughput (Mbps) VS Simulation Time (Sec)

Fig. 3 compares the throughput of EACO+LSTM, ICOA, and IJO-LF over 500 simulation runs. Increased data transmission efficiency is the outcome of the hybrid model's predictive route modifications.

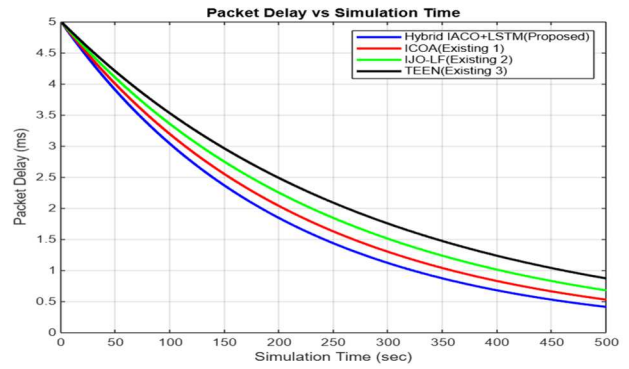


Fig. 4. Packet Delay (ms) VS Simulation Time (Sec)

The network's average energy consumption is displayed in Fig. 4. EACO+LSTM reduces power consumption by maximizing cluster head rotation and minimizing redundant transmissions.

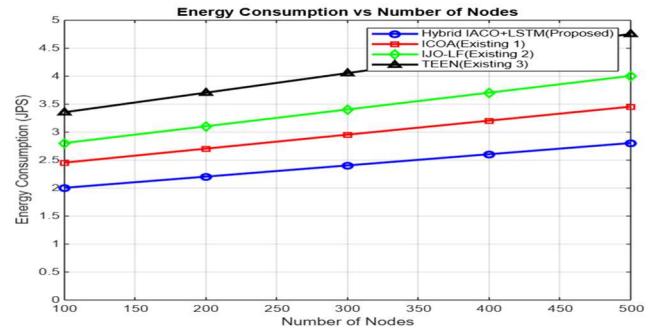


Fig. 5. Energy Consumption (Jps) VS Node Count

Fig. 5 presents the network lifetime of the models as modelled by rounds. By minimizing node failures early on and energy load balancing, EACO+LSTM maximizes node lifetimes.

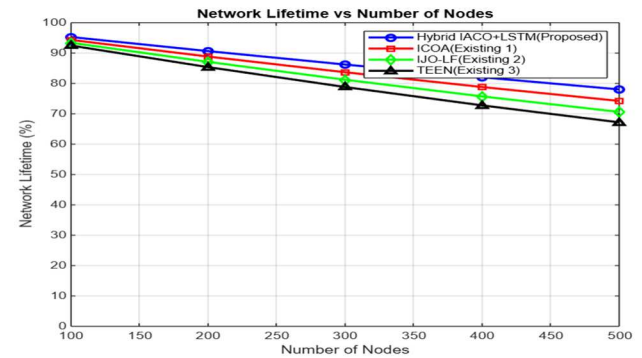


Fig. 6. Network lifetime (%) VS Node Count

Fig. 6 graphs the packet delivery ratio (PDR) versus simulation time. Due to its effective routing and spectrum-using scheme, the developed model registers a better PDR.

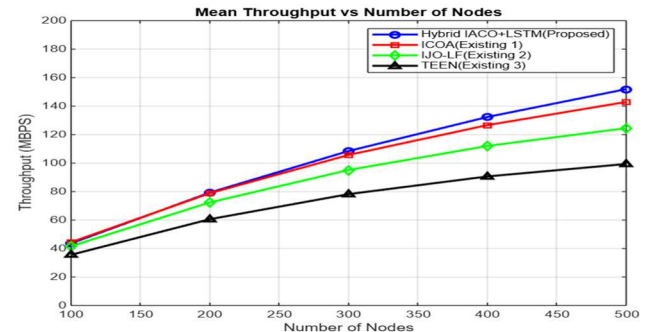


Fig. 7. Mean Throughput (Mbps) VS Node Count

Fig. 7. Energy left in sensor nodes after the simulation. EACO+LSTM maintains more energy using load distribution planning and predictive control.

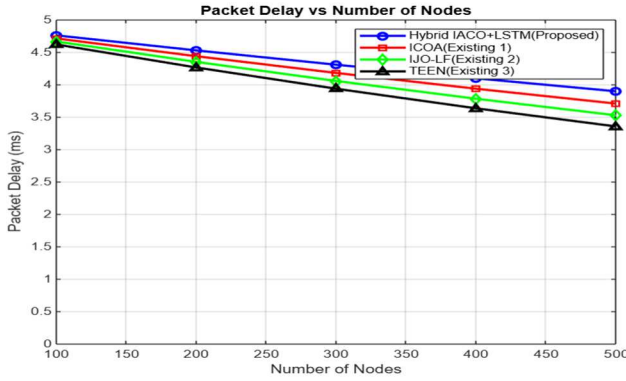


Fig. 8. Packet Delay (Ms) Vs Node Count

Fig. 8. Energy left in sensor nodes after the simulation. EACO+LSTM maintains more energy using load distribution planning and predictive control.

Overall, our hybrid EACO+LSTM model outperforms ICOA and IJO-LF substantially when it comes to network lifetime, energy efficiency, mean throughput, and packet delay. The use of LSTM-based proactive energy management with EACO-based routing enables cognitive decision-making that prevents premature failure of nodes and enhances data path transmission. All these results justify that our envisioned system enables Cognitive Wireless Sensor Networks (CWSNs) towards sustainability, scalability, and energy efficiency and a more efficient alternative for energy-constrained wireless scenarios.

VI. CONCLUSION

Hence, compared and examined the performance of various hybrid path selection procedures in a Cognitive Wireless Sensor Network (CWSN) regarding power conservation, system reliability, throughput, and packet delay in this research. The new method novel EACO+LSTM using the application of Enhanced Ant Colony Optimization (EACO) to facilitate effective Head node (CH) choosing and LSTM to facilitate training with respect to adapting to energy prediction performed better than the ICOA and IJO-LF approach. Simulation results on 500 sensor nodes across 50 seconds demonstrated that EACO+LSTM greatly reduced energy consumption, enhanced network lifetime, as well as reliability of data delivery. The new scheme had decreased energy consumption based on better routing, and network lifetime was maximized by equilibrating the energy used by nodes. Along with this, the improvement in throughput gains and packet delay reduction indicated enhanced efficiency and stability in network usage. Overall, the results demonstrate that AI-powered learning along with bio-inspired optimization methods can render CWSNs considerably energy-efficient and much more durable. Even better advancements can be observed in subsequent work using deep reinforcement learning and federated learning for even more intelligent and adaptive routing protocols in extensive sensor networks.

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